



United International University

Dept. of Electrical & Electronics Engineering

Course: Energy Conversion Laboratory

(Course Code: EEE206)

Project Report

DESIGN AND CONSTRUCTION OF A 100 VA SINGLE-PHASE STEP-DOWN TRANSFORMER (220V/22V)

Submitted By: Group - 03

No.	Name	ID
1.	Hasan Rayhan Rabbe	0212410026
2.	Sabbir Ahmed	0212410034
3.	Ashikur Rahman	0212410009
4.	Md.Mahbubur Rahman	0212410013
5.	Faysal Mahmud	0212410014
6.	Jannatun Mawa Simu	0212410012

AIM: TO MAKE A STEP DOWN TRANSFORMER

OBJECTIVE:

The main objective of this project is to design, construct, and test a 220V/22V step-down transformer using theoretical and practical knowledge of electrical machines. The project aims to calculate all necessary design parameters such as number of turns, current rating, and VA capacity, and then implement these calculations in a real transformer model. Another objective is to perform standard laboratory tests, including the open circuit test, short circuit test, and load test, in order to study the performance characteristics of the transformer.

ABSTRACT:

This project presents the design, construction, and experimental performance analysis of a singlephase 220V/22V step-down transformer. The objective of the work was to practically implement a transformer based on theoretical design principles and to evaluate its performance using standard testing methods such as no-load test, load test, rated load test, and short-circuit test. The experimental results were used to determine key performance parameters including voltage regulation and efficiency. The obtained results show that the constructed transformer operates with satisfactory efficiency and acceptable voltage regulation, validating the design methodology and construction process.

INTRODUCTION:

A transformer is a static electrical device that transfers electrical energy from one circuit to another through electromagnetic induction while maintaining the same frequency. Transformers are widely used in power systems to step up or step down voltage levels for efficient transmission and safe utilization of electrical energy. Step-down transformers are particularly important in lowvoltage applications such as power supplies, battery chargers, electronic devices, and control systems.

In this project, a single-phase step-down transformer rated at 220V/22V was designed and constructed. The main aim was to understand the practical aspects of transformer construction and to verify its theoretical performance through experimental testing. This project also helps in developing hands-on experience with transformer testing procedures and performance evaluation.

COMPONENT:

- Laminated iron core
- Bobbin (former)
- Enamel-coated copper wire (for primary winding)
- Enamel-coated copper wire (for secondary winding)
- Insulation paper / insulating tape
- Core clamps / screws
- Terminal block / connecting wires
- Varnish (optional, for noise and insulation)
- Multimeter (for testing)
- Load resistor / load bank (for load test)

Design and Construction:

The transformer was designed based on standard transformer design equations considering the supply voltage, desired output voltage, frequency, and estimated power rating. A laminated iron core was selected to reduce eddy current losses and improve magnetic performance. The primary winding was designed for 220V operation and wound first on the bobbin, followed by the secondary winding for 22V. Proper insulation was used between layers to ensure electrical safety and to prevent short circuits. After winding, the transformer was assembled with the core and mechanical support, and varnish was applied to improve insulation, reduce vibration, and enhance thermal stability.

Experimental Procedure:

To evaluate the performance of the transformer, several standard tests were performed. The no-load test was conducted by applying rated voltage to the primary while keeping the secondary open-circuited. This test was used to observe the secondary voltage and core behavior. The load test was carried out by connecting a 100-ohm resistive load to the secondary and measuring the corresponding voltages, currents, input power, and output power. A rated load test was also performed to approximate full-load conditions, where higher current was drawn from the secondary. The short-circuit test was conducted by shorting the secondary and applying a reduced voltage to the primary to obtain rated current, allowing the estimation of copper losses and equivalent impedance.

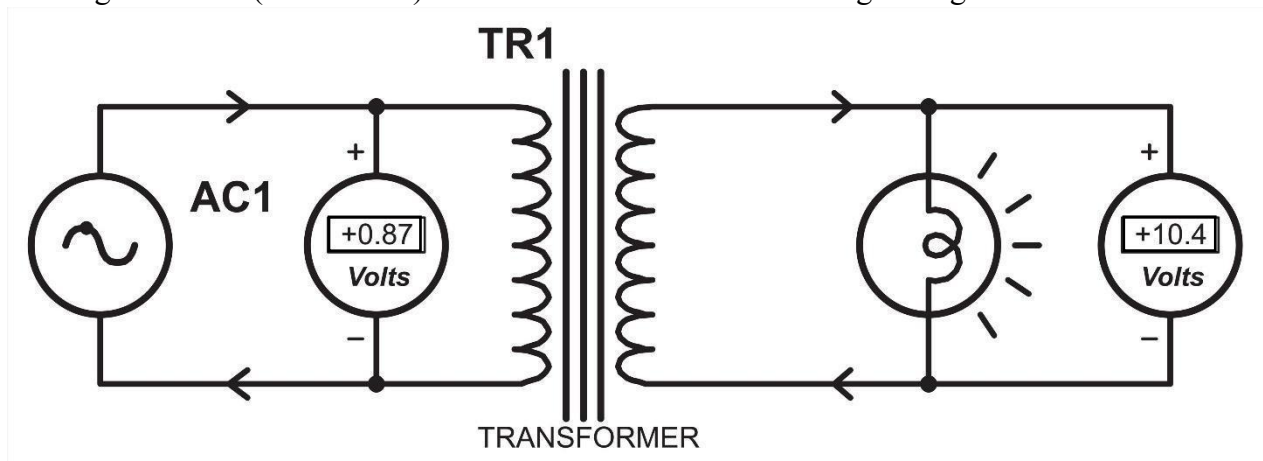
Test Results and Observations:

During the tests, the primary voltage V_p was maintained at **220 V ($\pm 5\%$)** to avoid over-fluxing the core. The primary and secondary currents were kept below their rated values ($I_p \leq 0.455\text{ A}$ and $I_s \leq 4.55\text{ A}$) to prevent excessive copper losses (I^2R).

During the short-circuit test, a reduced voltage of $V_{sc}=26\text{ V}$ was applied to ensure the short-circuit current $I_{sc}=4.5\text{ A}$ stayed within safe limits. Measuring instruments were selected such that their ranges exceeded the expected readings:

- **Voltmeter:** $V_{\text{meter}} \geq 250\text{ V}$
- **Ammeter:** $I_{\text{meter}} \geq 5\text{ A}$
- **Wattmeter:** $P_{\text{meter}} \geq 100\text{ W}$

The insulation resistance was maintained at a high level ($R_{\text{ins}} \gg 1\text{ M}\Omega$) to prevent leakage currents. Additionally, the temperature rise was monitored to ensure that any increase in winding resistance (where $R \propto T$) remained within safe limits during testing.

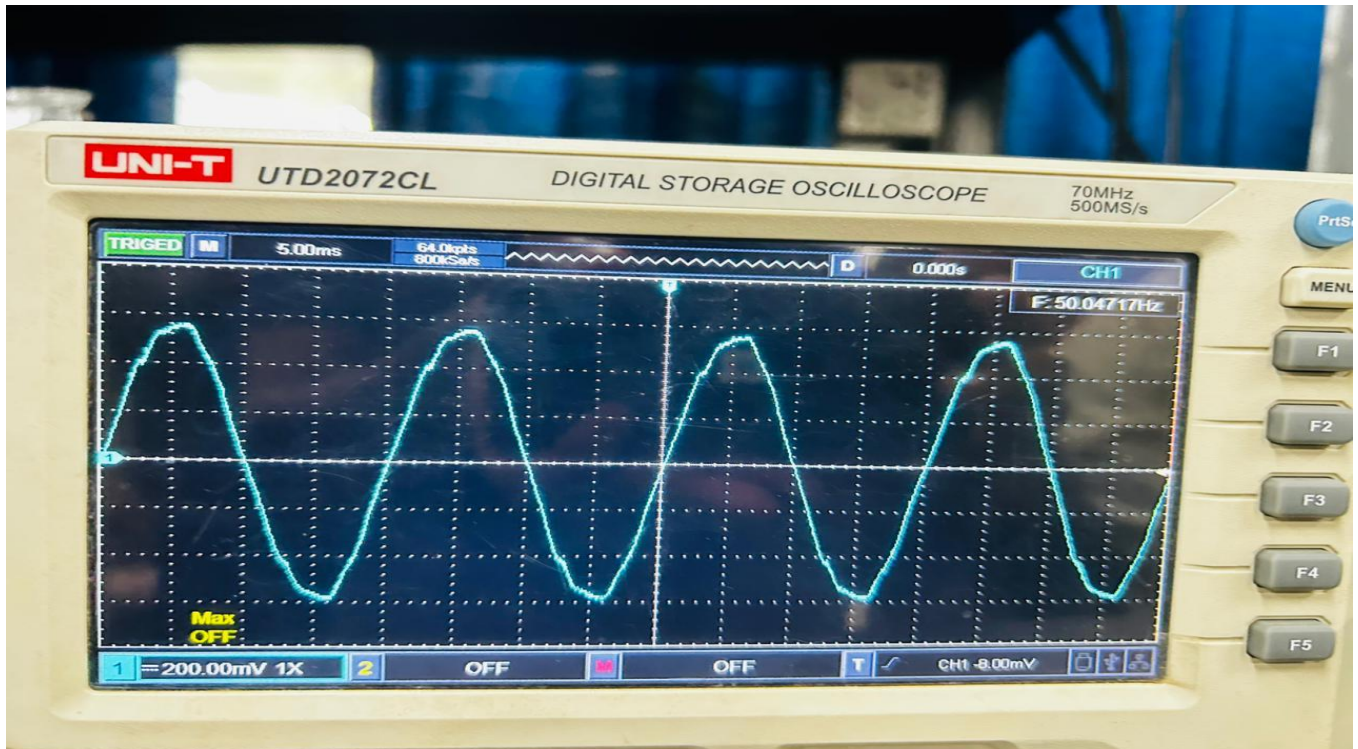


Transformer Specifications:

A transformer is a static electrical device that transfers energy via electromagnetic induction. By varying the number of turns between the primary and secondary windings, the voltage is "stepped down" while the frequency remains constant.

- Primary Voltage: 220 V AC
- Secondary Voltage: 22V AC
- Frequency: 50Hz
- Type: Step-down Transformer
- Core Type: Laminated Iron Core
- Cooling: Air Natural

Output ac signal:



Construction Details:

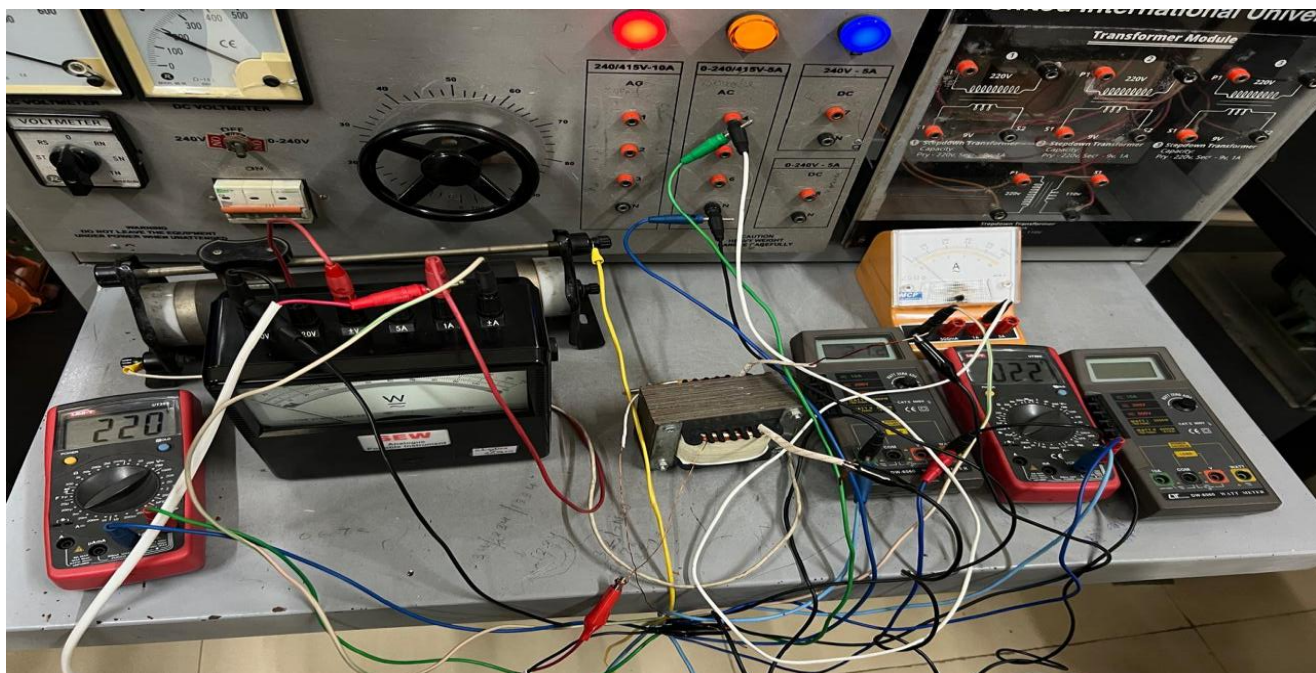
Transformers use a Laminated Iron Core to provide a low-reluctance path for magnetic flux. Laminations are used specifically to break up the path of eddy currents, which reduces heat loss.

- Windings: The primary (high voltage) was wound first, with the secondary (low voltage) on top. This minimizes leakage flux, ensuring that most of the magnetic field produced by the primary reaches the secondary.
- Protection: Varnish and insulation were applied to prevent "turn-to-turn" short circuits and reduce audible humming caused by magnetostriction.

No-Load Test:

This test is performed with the secondary open to measure Core Losses (Hysteresis and Eddy Current losses). Since the secondary current is zero, the power consumed is almost entirely due to the magnetic properties of the core.

- Measured V_p : 220V
- Measured V_s : 22v



Rated Load Test:

This test determines how the transformer performs at its design limit. As current increases, the Copper Losses I^2R increase, which generates heat and causes the output voltage to "sag."

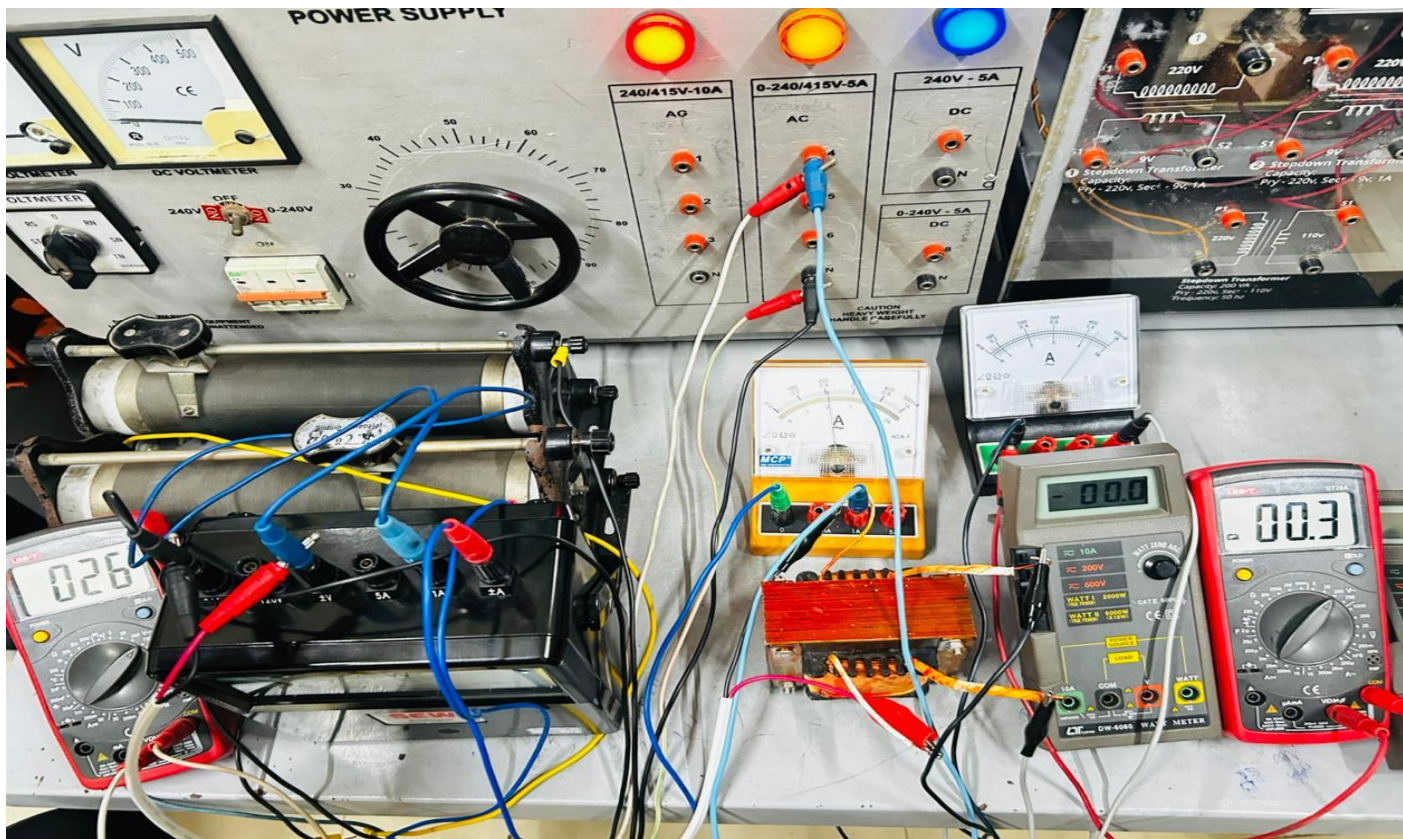
- Data: $V_p=220\text{ V}$, $V_s=20\text{ V}$, $I_p=0.4\text{ A}$, $I_s=4.35\text{ A}$.
- Power: $P_{in}=70\text{ W}$, $P_{out}=66\text{ W}$.
- Observation: The drop to 2 V is the result of the internal impedance of the windings under full load.



Short-Circuit Test:

By shorting the secondary and applying a low voltage V_{sc} , we can measure the Total Copper Loss at rated current without the core losses interfering.

- Data: $V_{sc}=26\text{ V}$, $I_{sc}=0.42\text{ A}$
- Purpose: This test is essential for calculating the Equivalent Impedance and predicting how the transformer will behave under a fault condition.



Performance Metrics:

- **Voltage Regulation**

Regulation indicates the transformer's ability to maintain a constant voltage regardless of the load.

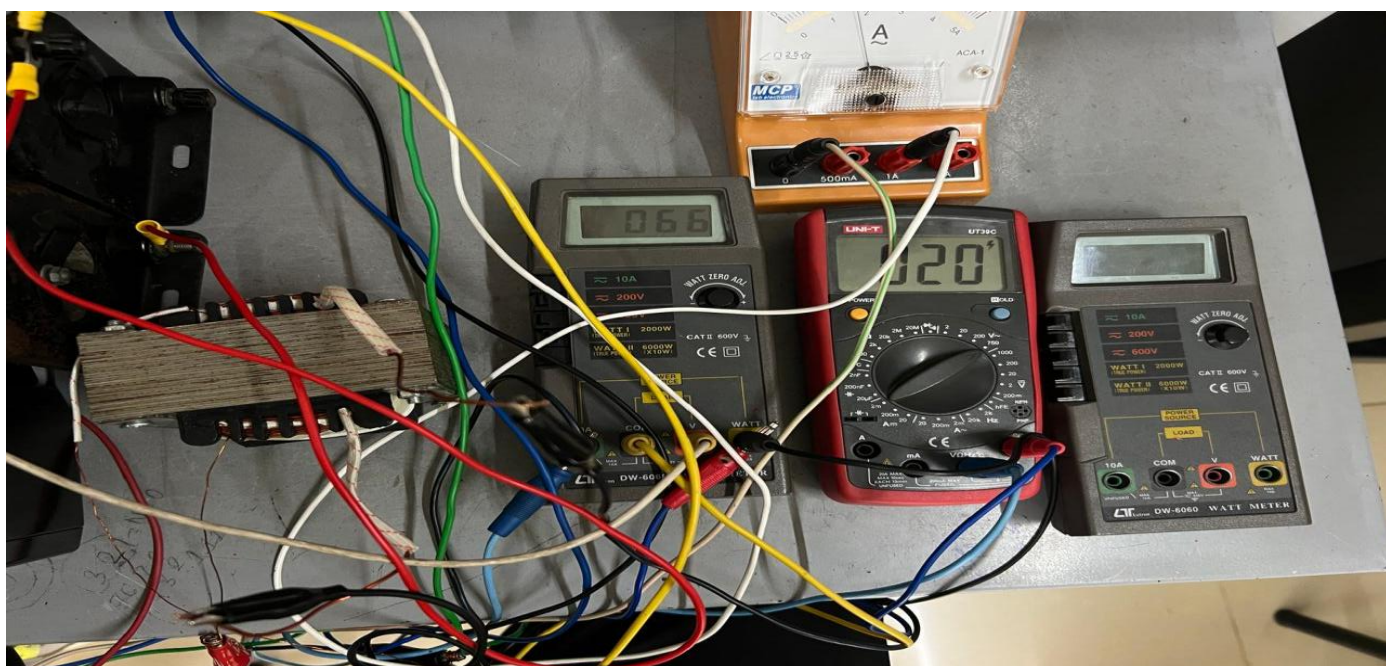
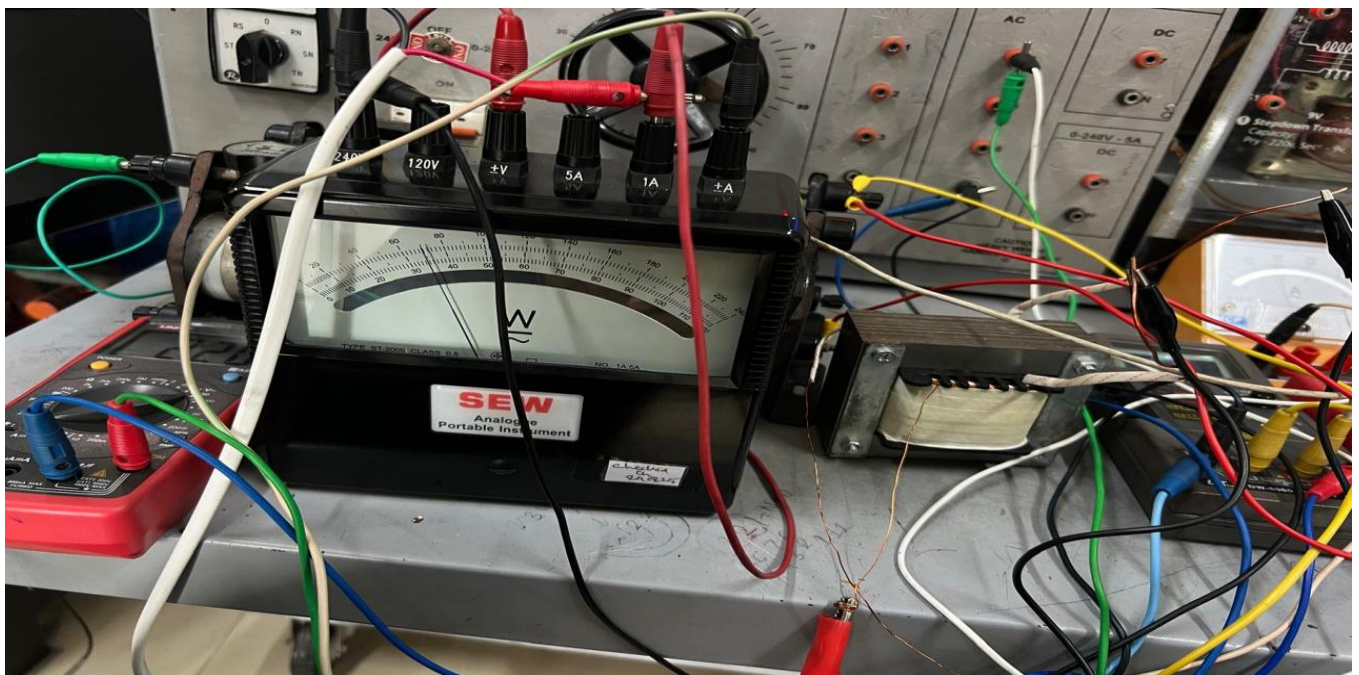
A lower percentage is generally better.

- **Calculation:** $(22-20)/20 \times 100 \approx 10\%$
- **Result:** An 10% drop is acceptable for small-scale power applications.

Efficiency (η) :

Efficiency is the ratio of useful work to the total energy supplied. It accounts for both magnetic (core) and electrical (copper) losses.

- Calculation: $(66/70) \times 100 \approx 94.3\%$
- Result: The 5.7% loss is dissipated as heat, which is typical for a transformer of this size and cooling type.



CONCLUSION:

In this project, a 220V/22V single-phase step-down transformer was successfully designed, constructed, and tested. Standard transformer tests including no-load test, load test, rated load test, and short-circuit test were performed to analyze its performance. The voltage regulation was found to be approximately 10% and the efficiency was approximately 94% at rated load. These results indicate that the transformer operates efficiently with acceptable voltage drop and meets the expected performance criteria. This project effectively demonstrates the practical implementation of transformer theory and provides valuable hands-on experience in electrical machine design and testing.

PRECAUTIONS:

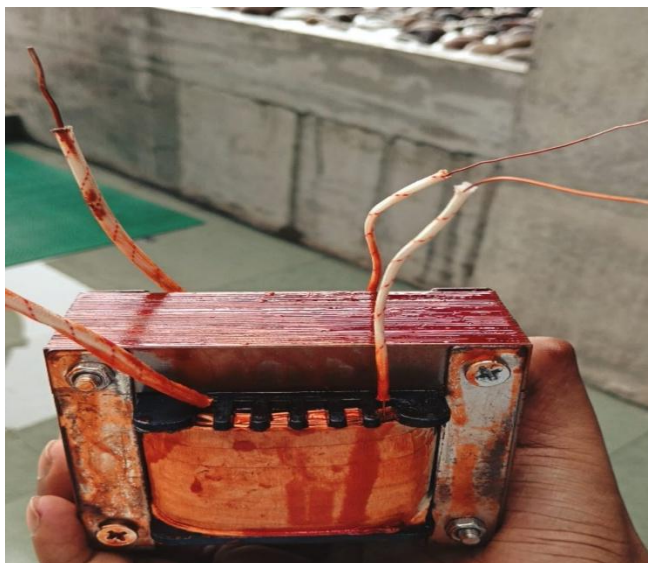
During the experiment, proper insulation was ensured between primary and secondary windings to prevent short circuits and electrical hazards. All connections were made tightly and securely to avoid loose contacts and overheating. The supply voltage was applied gradually using a variac to prevent sudden inrush current. Measuring instruments with suitable ranges were used for accurate readings. During the short-circuit test, only reduced voltage was applied to limit current and prevent damage. The transformer was not operated beyond its rated limits and was allowed to cool between tests to avoid overheating.

Final Discussion & Applications:

Real-world transformers face limitations like magnetic leakage and winding resistance. Our tests confirm these theories, as seen in the voltage drop and efficiency figures.

Applications: Ideal for power supplies, battery chargers, and control circuits.

Limitations: Natural air cooling limits the power density, and manual winding can lead to slight imbalances in the magnetic flux.



----- **THANK YOU** -----